

Lecture 4 - Proportional Control

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4.1 Proportional Control

- Analyze a control system with P control: Compare Open & Closed loop
- Look at:
 1. Tracking
 2. Regulation (disturbance rejection)

Example: Cruise control on a bumpy road

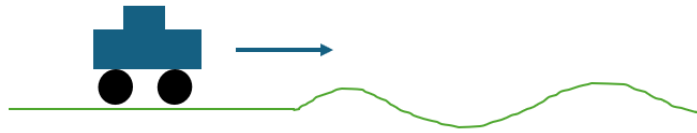


Figure 4.1: Cruise control on bumpy road.

First, we choose a model and convert it into a block diagram, then add control.

- Model for speed $y(t)$ of the car:

$$\underbrace{\dot{y}(t)}_{\text{accel}} + ay(t) = bu(t) + w(t), \quad \text{assume: } a, b > 0 \text{ and } y(0) = \dot{y}(0) = 0$$

Where w is the disturbance due to the change in slope.

- Take Laplace

$$sY(s) + aY(s) = bU(s) + W(s)$$
$$Y(s) = \underbrace{\frac{b}{s+a}}_{G_{UY}} U(s) + \underbrace{\frac{1}{s+a}}_{G_{WY}} W(s)$$

- Block Diagram

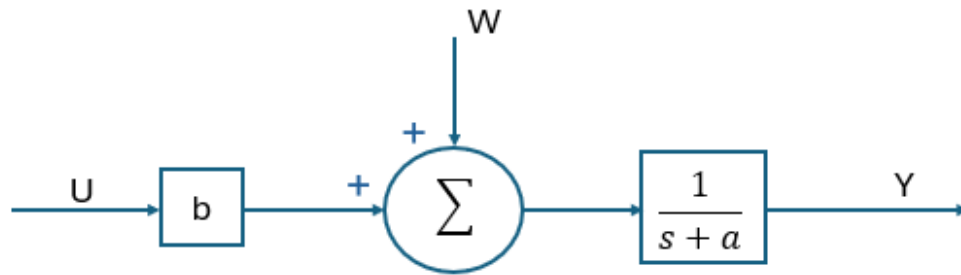


Figure 4.2: System dynamics only, we have not added control yet.

- Control Objective

Design a controller H that makes the output Y track the reference r (e.g. get y to track $r = 60$ mph).

4.1.1 Open Loop Proportional Control

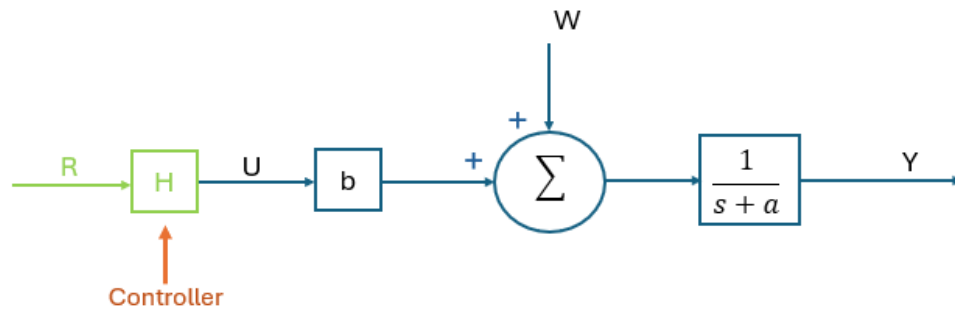


Figure 4.3: Open loop proportional control block diagram.

Find transfer functions G_{RY} and G_{WY} :

$$G_{RY} = \frac{Hb}{s+a} = H \cdot G_{UY}$$

$$G_{WY} = \frac{1}{s+a} = G_{WY}$$

- Design H s.t. whenever $r(t) = r$ (step input $R(s) = \frac{r}{s}$) and $w(t) = 0$
 $y(t) \rightarrow r$ as $t \rightarrow \infty$ (steady state: $s = 0$)

→ Final Value Theorem (FVT)

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s) = s \underbrace{G_{RY}(s)R(s)}_{Y(s)} = s \cdot \frac{Hb}{s+a} \cdot \frac{r}{s} = \frac{Hb}{a} \cdot r$$

$$\text{For } y(t) = r \text{ as } t \rightarrow \infty \implies \frac{Hb}{a} = 1 \implies H = \frac{a}{b}$$

However, notice H is based on our plant model via "a".

Our model must be perfect for this to work!

→ Substituting H into our transfer functions:

$$\left. G_{RY} = \frac{a}{s+a} \right\} \text{DC gain (s=0) is } \frac{a}{a} = 1, \text{ as intended.}$$

$$\left. G_{WY} = \frac{1}{s+a} \text{ (this hasn't changed! Controller didn't affect it.)} \right\} \text{DC gain} = \frac{1}{a}$$

For step input r, and step disturbance W:

$$\lim_{s \rightarrow 0} sY(s) = 1r + \frac{1}{a}W$$

$$\implies \text{Perfect } \underline{\text{tracking}} \text{ (} Y = r \text{)}$$

$$\implies \text{Great } \underline{\text{regulation}} \text{ if: } \begin{cases} W \text{ is small} \\ a \text{ is large} \end{cases}$$

What about sinusoidal inputs - inputs that change quickly?

- To study tracking & regulation over different frequencies, we look at gain on Bode plots (focusing on magnitude, ignoring phase).

- * Good tracking: $|G_{RY}| \approx 1$

- * Good regulation: $|G_{WY}| \ll 1$

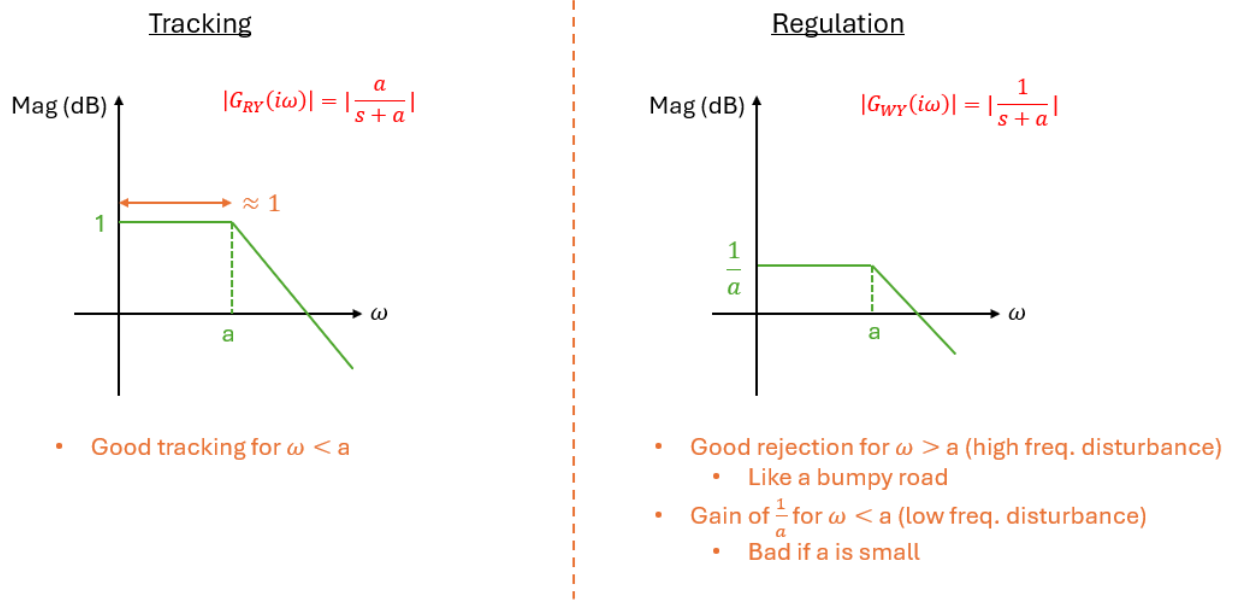


Figure 4.4: Tracking and regulation for open loop proportional control.

4.1.2 Closed Loop Proportional Control

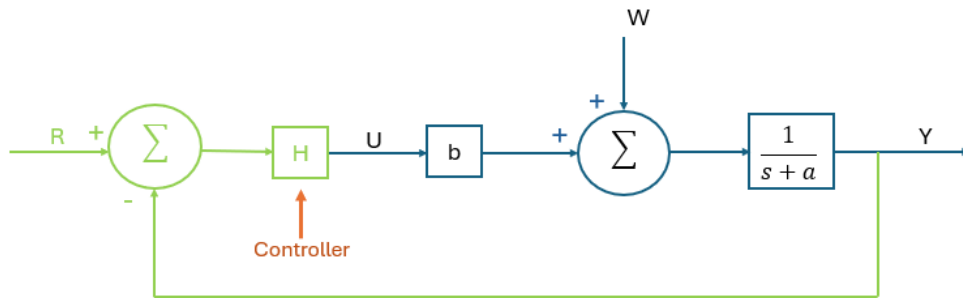


Figure 4.5: Block diagram for closed loop proportional control.

Find transfer functions:

$$G_{RY}(s) = \frac{\frac{Hb}{s+a}}{1 + \frac{Hb}{s+a}} = \frac{Hb}{s+a+Hb} \Rightarrow G_{RY}(0) = \frac{Hb}{a+Hb}$$

$$G_{WY}(s) = \frac{\frac{1}{s+a}}{1 + \frac{Hb}{s+a}} = \frac{1}{s+a+Hb} \Rightarrow G_{WY}(0) = \frac{1}{a+Hb}$$

Look at poles to check stability (H is just a scalar in P control, but now in closed loop it appears in the denominator).

$$s + a + Hb \Rightarrow s = -a - Hb$$

→ Pole is guaranteed to be in the LHP (negative) if $H > 0$ (recall we assumed $a, b > 0$).

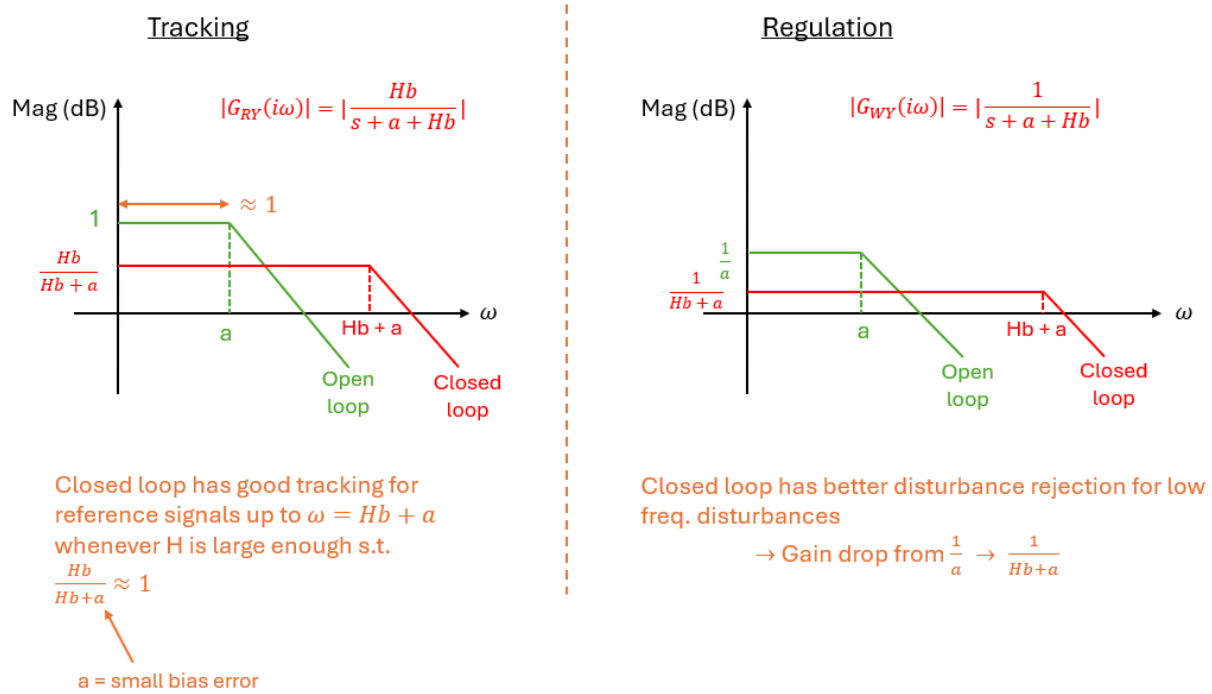


Figure 4.6: Tracking and regulation for closed loop proportional control.

∴ Increasing H improves tracking and disturbance rejection. However, physically, very large H is not possible because:

1. Actuator saturation
2. System can become unstable due to unmodeled dynamics

$$\text{e.g. } G_{RY} = \frac{1}{s+a} \cdot \underbrace{\frac{1}{\left(\frac{s}{10} + 1\right)^2}}_{\text{unmodeled dynamics}} \rightarrow \text{More on this later}$$